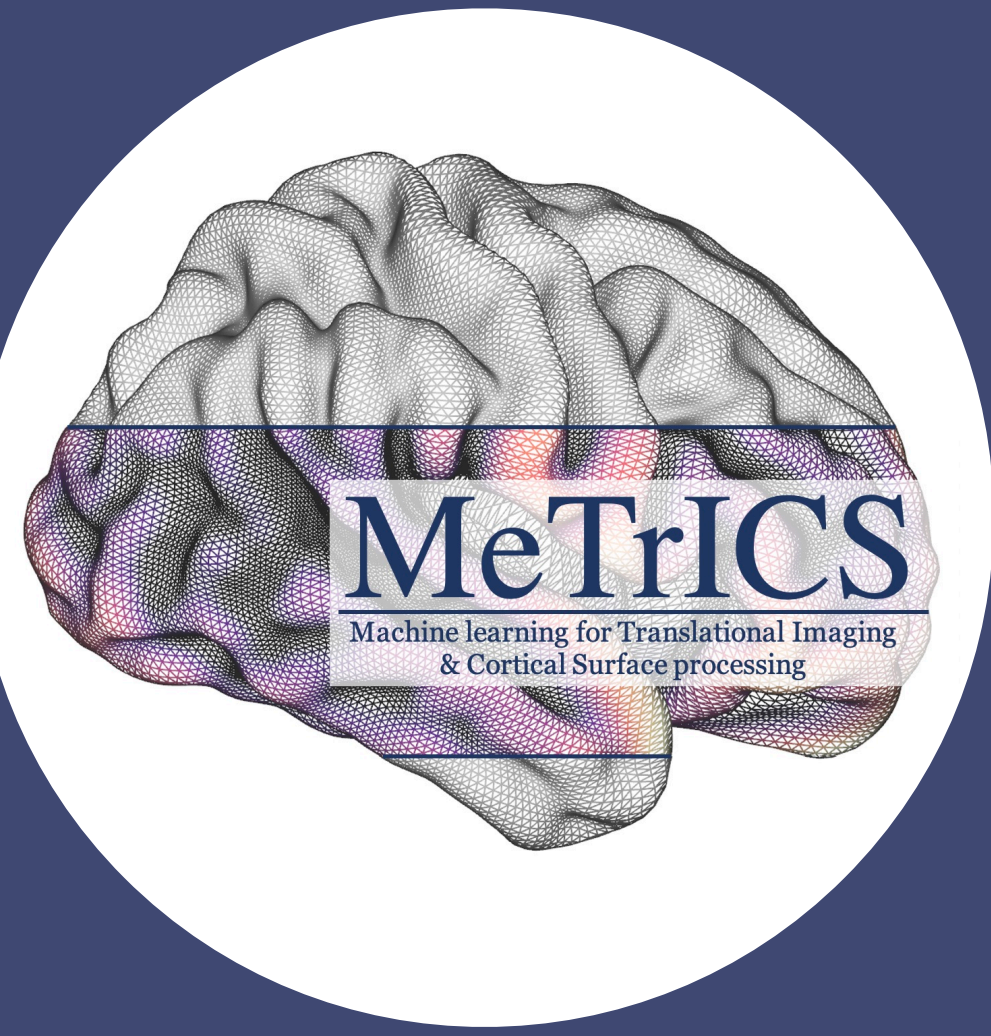




Shape-aware normative modelling reveals distinct cortical developmental pathways

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Summary

- Cortical shape context matters. Normative models that respect gyral and sulcal anatomy produce tighter developmental trajectories – and a more sensitive reference for detecting atypical or disease-related patterns.**

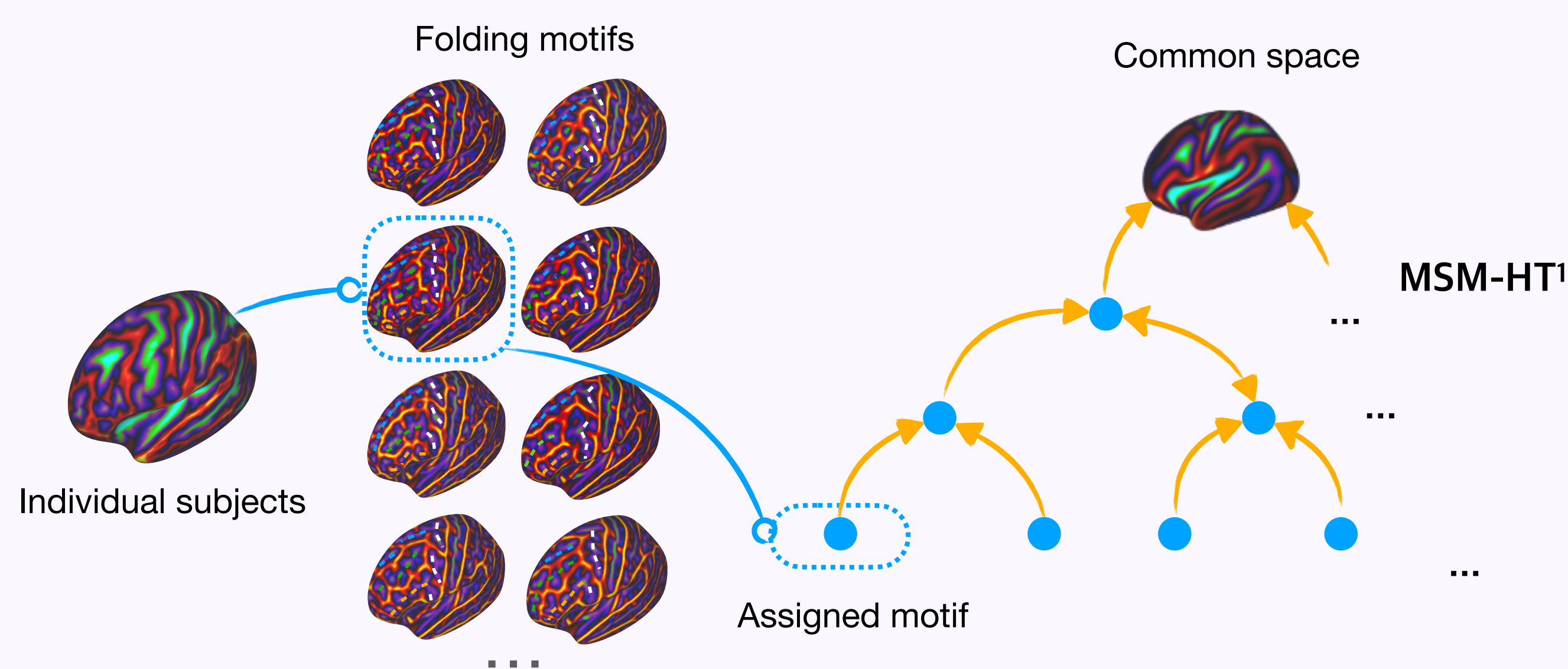
Motivation

- Conventional registration aligns everyone to a single average template – blending gyral and sulcal tissues across individuals. This blurs the very developmental signal normative models aim to capture.

Method

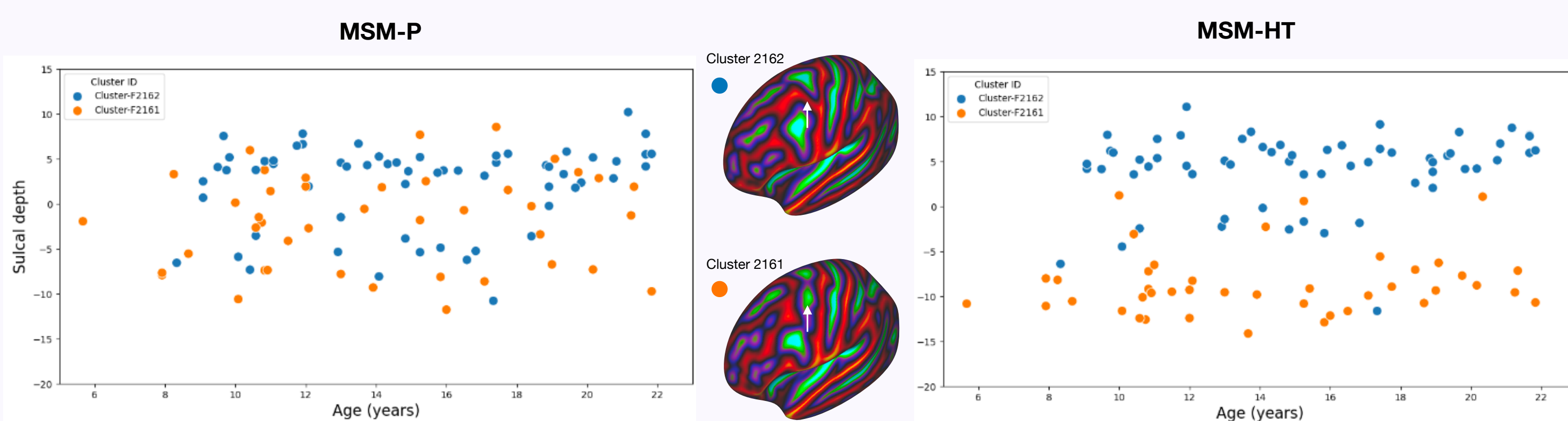
Address registration

- Apply MSM-HT¹ to group subjects by folding similarity and align them to a common shape-aware space.



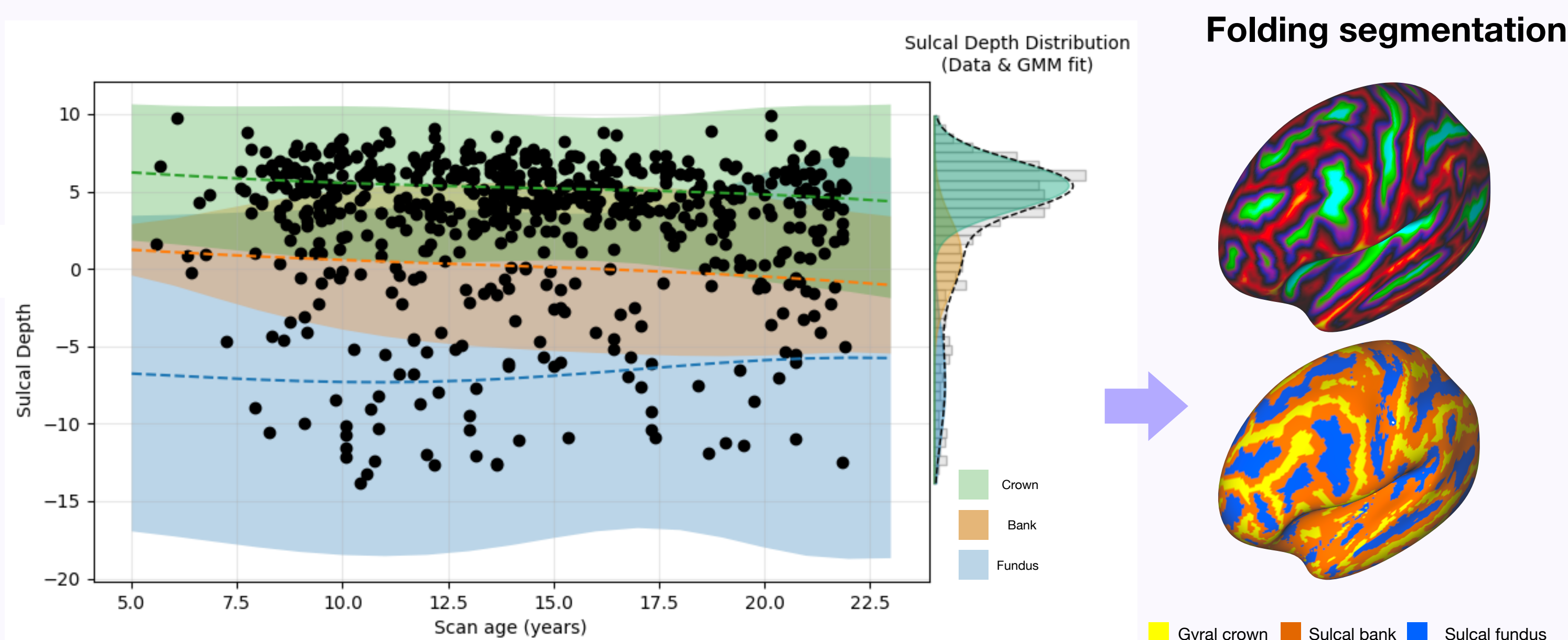
Separate sulci/gyri trajectory

- MSM-HT alignment clusters vertices by tissue type, cleanly separating gyral and sulcal developmental trajectories.



Model distinct trajectories (MoGPs)

- Distinct developmental pathways are modelled vertex-wise using Mixture of Gaussian Processes (MoGP).

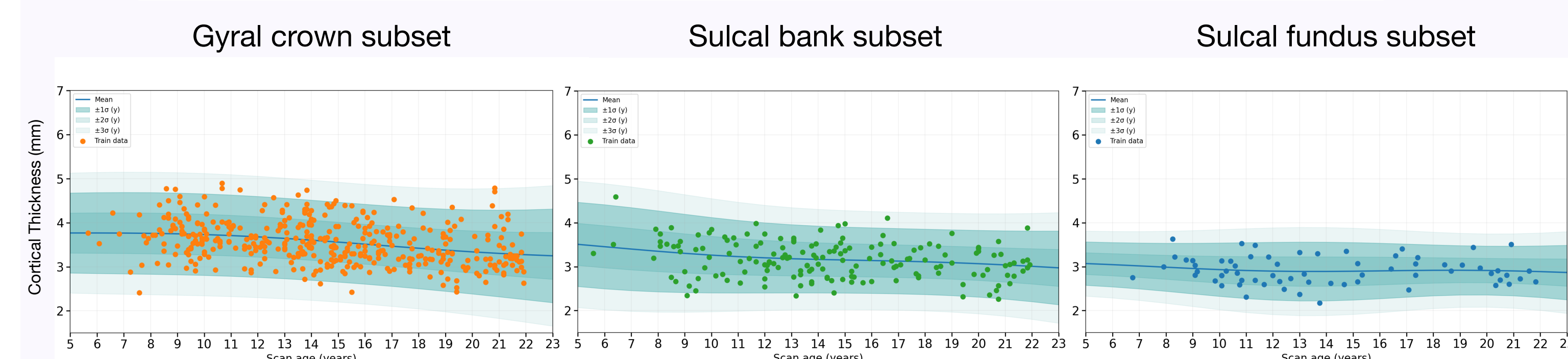
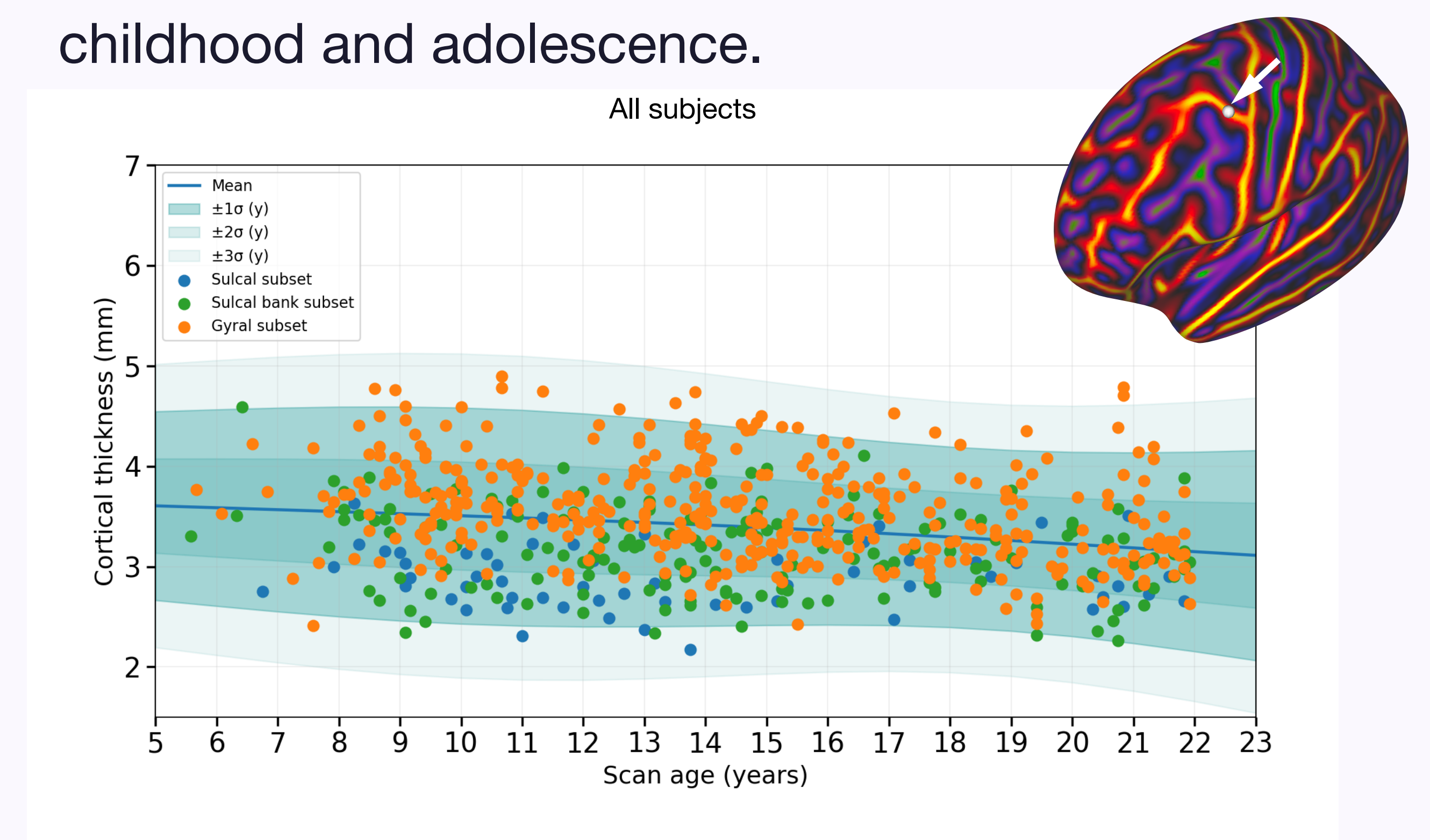


► 652 subjects aged 5 to 21 years (351 females) from the Lifespan Human Connectome Project-Development (HCP-D²).

Results

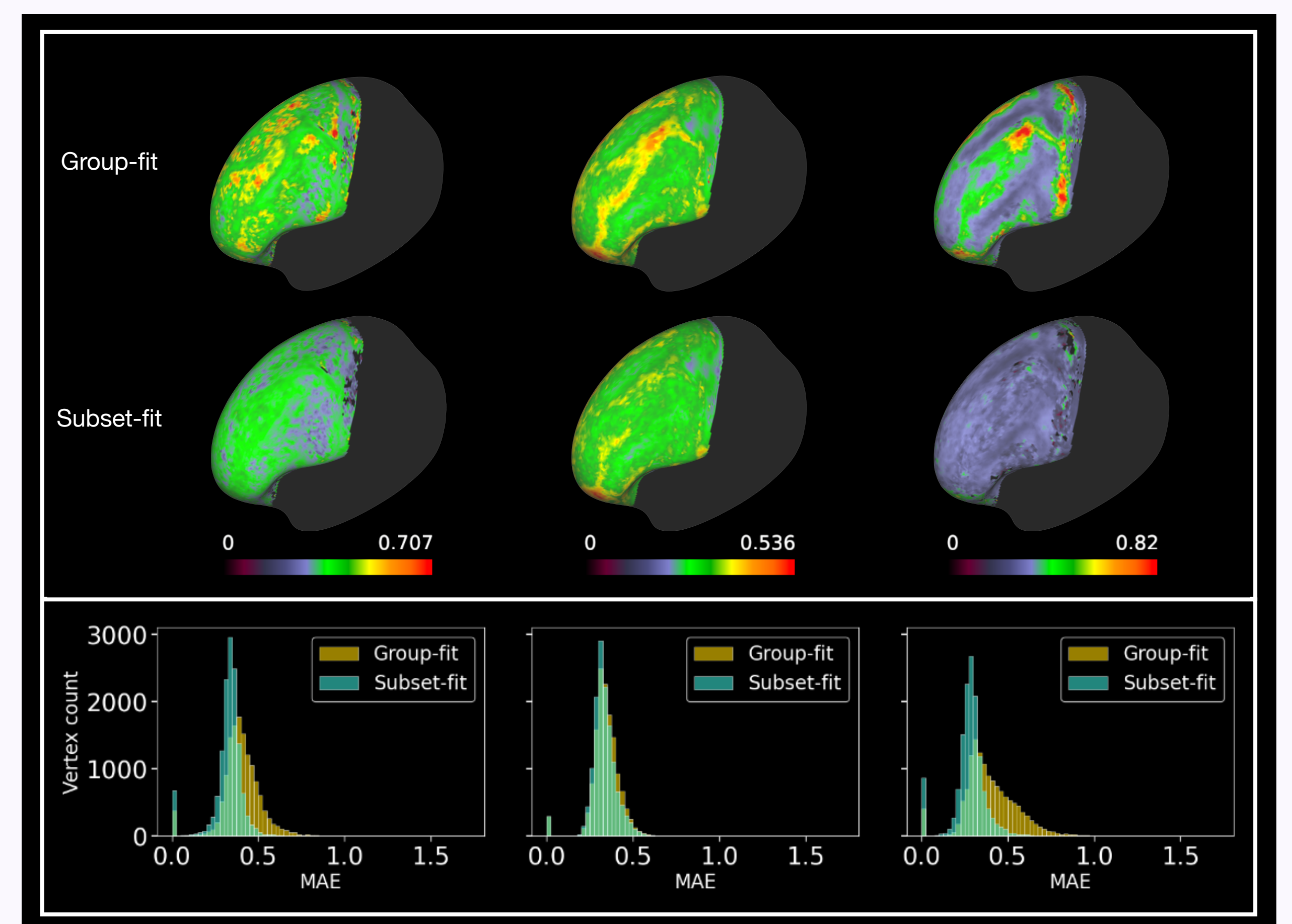
Cortical thickness by folding component

- Cortical thickness was consistently greatest for gyral crowns and lowest for sulcal fundi, with all three components showing age-related thinning across childhood and adolescence.



Shape-aware models outperform baseline

- Component-specific GP models consistently achieved lower MAE than baseline models that ignore folding tissue, demonstrating that shape context directly improves normative accuracy.



This framework provides a biologically grounded reference for typical cortical development, with the potential to increase sensitivity to subtle deviations in clinical populations.

References

- Guo et al. "Motifs of brain cortical folding from birth to adulthood." bioRxiv (2025) yourong.2.guo@kcl.ac.uk
- Somerville et al. "The Lifespan Human Connectome Project in Development." NeuroImage (2018)

